HARSHINI JEGANNATH

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EDUCATION

The University of Chicago **Sep 2025 – Dec 2026 (Expected)**
Master of Science in Computer Science, GPA-3.78/4 *Chicago, Illinois, USA*
Coursework: Algorithms, Databases, Applied Data Analysis, Generative AI, Parallel Programming, Web and Cloud Development, GPU

Vellore Institute of Technology **Aug 2021 - Aug 2025**
Bachelor of Science in Computer Science, GPA-9.63/10 *Chennai, Tamil Nadu, India*
Coursework: Data Structures, Networks, Machine Learning, Software Engineering, AI, Computer Architecture, Operating Systems
Awards: Meritorious Award (Department Top 10), Attendance Award
Conferences: IEEE - International Conference on Data Science & Business Systems (2025)

EXPERIENCE

FLSmidth **July 2025**
Machine Learning Engineer Intern *Chennai, India*
- Developed predictive maintenance ML pipelines using Python and scikit-learn, processing 10,000+ IoT sensor data points daily for anomaly detection and equipment failure prediction.
- Built end-to-end data ingestion and feature engineering pipelines on Microsoft Azure, implementing data preprocessing and transformation for real-time ML model inference.
- Designed containerized ML model deployment architecture using Docker, optimizing model serving latency by 25% for edge-to-cloud predictive analytics systems.

Samsung R&D Institute India **Oct 2023 - Mar 2024**
Research Internship *Chennai, India*
- Developed deep learning models for computer vision text-style detection using Python, TensorFlow, and OpenCV, improving OCR system accuracy for identifying bold, italic, underlined, and strikethrough text.
- Engineered automated data pipeline generating 12,000+ labeled training samples using Python and PIL, implementing data augmentation techniques for robust deep learning model training.
- Implemented OpenCV pipelines for underline and strikethrough detection, CNN for italic classification, and KNN for bold detection, achieving 98% accuracy.
- Deployed trained ML models into production for real-time inference, reducing text-style detection latency by 40% and enhancing end-to-end OCR system performance.

Ethnus Codemithra **Aug 2023 - Nov 2023**
Internship *Chennai, India*
- Architected and developed full-stack carpooling web application using Python, JavaScript, and AWS, implementing responsive UI and RESTful API backend for user management and ride scheduling.
- Deployed scalable backend infrastructure using AWS EC2, RDS, and Lambda, implementing serverless functions for automated user-group matching algorithm, improving ride allocation efficiency by 30% for 200+ active users.
- Integrated AWS S3 for secure file storage and implemented notification microservices using SES and SNS for real-time email and push notifications, increasing user engagement by 25%.

TECHNICAL SKILLS

Languages and Frameworks: Python, C, C++, Java, R, SQL, HTML, CSS, PHP, JavaScript, MATLAB, Pandas, NumPy, SQL, MongoDB, PostgreSQL, golang, NumPy, seaborn, Matplotlib, scikit-learn, Keras, PyTorch, CUDA, LLM, RAG
Tools and Infrastructure: Microsoft Azure, AWS, Google Cloud Platform, Tableau, Power BI, Jupyter Notebook, Git, Microsoft Suite, Eclipse, VMware, Linux, Windows, macOS, Google Colab, Google Suites, Excel.
Publications: "Inclusivity and Accessibility Focused Text Summarization"
Certificates: AWS Certified Solutions Architect, Microsoft Azure AI, Data Science, Data Analytics, Ethical Hacking.

PROJECTS

Privacy-Preserving HR Chatbot with Federated Learning | *Python, NLP, Federated Learning, TF-IDF, LIME, Explainable AI*
- Built NLP chatbot using Federated Learning for distributed model training, implementing TF-IDF vectorization, cosine similarity matching, and LIME for explainable AI predictions with privacy-preserving architecture.

AI Music Mood Curator – Agentic RAG Playlist System | *Python, Claude API, LangGraph, RAG, MCP, Prompt Engineering, Streamlit*
- Built a multi-agent RAG pipeline with LangGraph (mood analysis, semantic retrieval, LLM playlist curation), using structured JSON-schema prompts to the Claude API and grounding outputs in a FAISS index of 5,000+ song embeddings.
- Exposed the pipeline as an MCP server enabling Claude Desktop to autonomously call retrieval and generation tools; added Responsible AI controls (diversity filtering, per-track explainability) to reduce popularity bias and filter bubbles.

Inclusive - Accessible Text Summarization using Transformers | *PyTorch, T5, Hugging Face, HAN, ROUGE, NLP, Deep Learning*
- Developed a generative NLP summarization system using a pretrained T5 transformer and Hierarchical Attention Networks (HAN) with PyTorch, integrating text-to-speech, translation, and an accessible UI.